

TEXT ANALYTICS

Introduction

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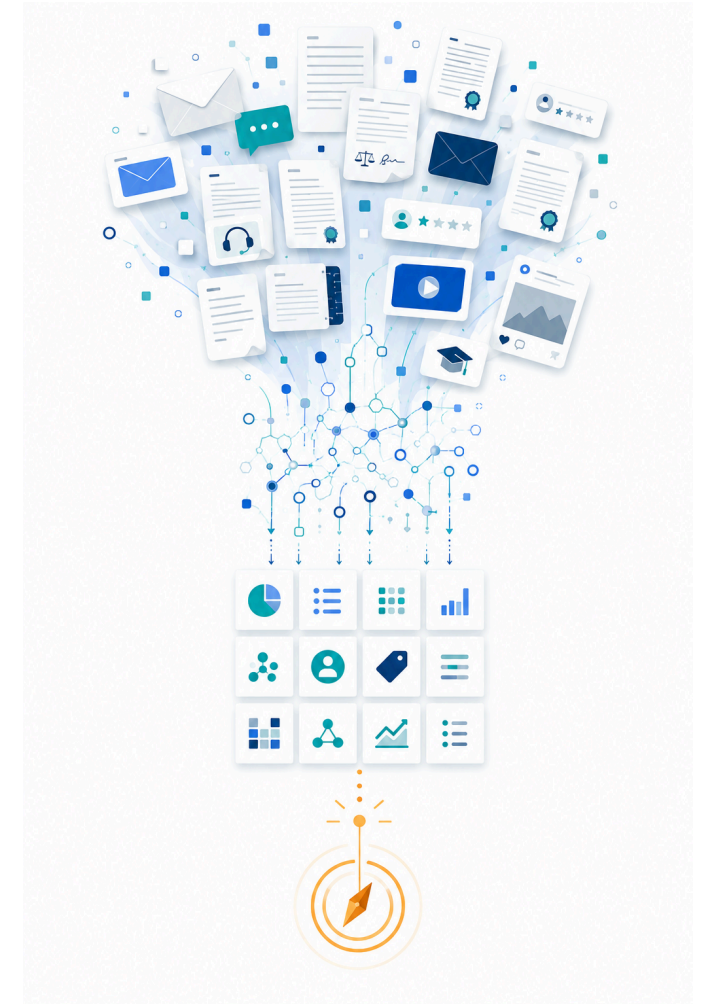
What Is Text Analytics?

The Data We Are Talking About

Most of the data an organization holds is not in a table, but in natural language: emails, tickets, contracts, reviews, papers, chat logs, posts...

- It has **no schema**: no columns, no types, no constraints.
- It is **written for humans**, by humans, and assumes everything humans assume.
- There is a great deal of it, and it arrives continuously.

Text analytics is the business of turning that into something you can compute with, and then into something you can act on.



Text analytics encompasses several relevant tasks:

Text classification. Given a text and a set of categories, choose one: sentiment and emotion, hate speech and abuse, topic, spam.

Question answering. Answer questions over a collection. Difficulty varies enormously:

- *"What is the capital of France?"*
- *"How many years have passed since Dante's birth?"*
- *"What were the social effects of the pandemic?"*

Text Analytics Tasks

Fact-checking and fake news detection. Decide whether a claim can be verified against sources, or learn the features that separate real from fabricated reporting.

Summarization. *Extractive* selects sentences from the original; *abstractive* generates new text.

Machine translation. Convert between languages, with all the ambiguity that entails

Generation. From a prompt, produce coherent text. Chatbots, assistants, drafting, summarising.

Why Natural Language Is Hard

The use of a complex language is a distinctive trait of humans, and every property that makes it expressive makes it difficult to process.

- Thousands of symbols, complex syntax
- Potentially **ambiguous** in part of speech, word sense and structure
- Relies on **shared world knowledge**:
"I'm faster, like Kimi Antonelli"
- Learned from scratch over a lifetime
- **Unstructured**, with high variability
- Carries **implicit** content: what is meant exceeds what is written
- Rich in entities, events and relations, inside and outside the text
- **Evolves**: *"I googled it", "it's in the cloud", "crowdfunded"*

Ambiguity Is Not an Edge Case

It is the normal condition of language. Every level of analysis has its own.

level	example	the problem
word sense	<i>I went to the bank</i>	river or financial
part of speech	<i>time flies like an arrow</i>	verb or noun
attachment	<i>I saw the man with the telescope</i>	who has the telescope
reference	<i>The trophy did not fit in the suitcase because it was too big</i>	trophy or suitcase
intent	<i>"Great, another meeting."</i>	enthusiasm or sarcasm

A human resolves all without noticing. Every one of them is a research area.

Natural Language Processing (NLP) is the branch of Artificial Intelligence that studies algorithms for processing written and spoken natural language.

It is genuinely interdisciplinary, involving computer scientists, linguists and cognitive scientists, and it exists because of a simple mismatch:

- there are **massive** amounts of text and speech,
- humans **cannot** read them at that scale,
- and the information inside them is **worth having**.

Text Mining and Text Analytics

The two terms are often used interchangeably. The distinction worth keeping is one of **purpose**.

Text Mining

Extracting patterns and knowledge *from* text.

- Classification, clustering, topic modeling, entity recognition
- Fake news detection, fact-checking, translation
- The output is a **model** or a **label**

Text Analytics

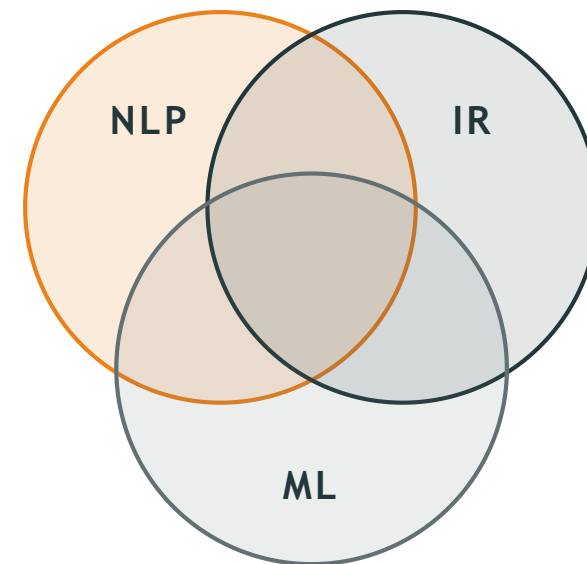
Interpreting those outputs to *decide something*.

- Trends, opinions, decision support
- Sentiment over customer feedback, movement of a debate over time
- The output is an **insight**

Three Interlocked Disciplines

Text analytics is built by combining three fields:

- **Natural Language Processing:** recognizing and processing linguistic elements and structures.
- **Information Retrieval:** the statistical properties of language, and finding things in collections.
- **Machine Learning:** statistical and neural methods that build descriptive and predictive models.



They are not separable.

- NLP methods are built on top of IR and ML;
- ML borrows IR's measures to define its objectives;
- ML assumes text that NLP and IR have already made tractable.

Where You Meet This Already

Everything in this course is running somewhere you used it this week.

you use it as	it is built from
search that understands a question	an inverted index, weighting, ranking
a spam folder that is usually right	features, a classifier, an honest evaluation
autocomplete in a message	next-word prediction from counts
a translation that reads naturally	attention over a sequence
a trending-topics panel	tokenization, normalization, association
a chatbot answering to your question	(probably) all of the above + generation

Understanding and Generation

Natural Language processing can be further divided into:

Natural Language Understanding (NLU)

Interpreting language by extracting structure and meaning from content.

- Analyzes aspects of language
- Maps input into a **representation** a machine can use

Natural Language Generation (NLG)

Producing meaningful language from data. Classically in three stages:

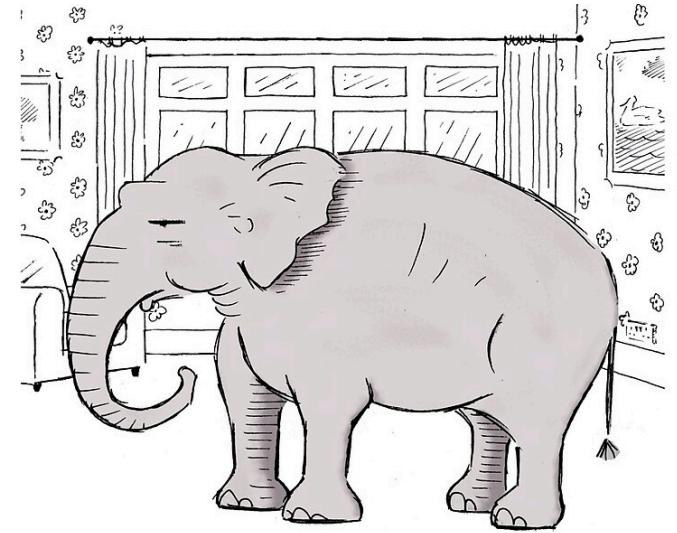
1. **Text planning:** what to say
2. **Sentence planning:** how to phrase it
3. **Text realisation:** producing text

Is Generative AI Absorbing Text Analytics?

Nowadays, when talking about text analytics one thinks immediately about **Generative AI**, i.e., the task of generating text. Generative AI sits across NLP, IR and ML, and it changed what each is used for.

- It absorbed a great deal of **NLP**: tokenization, tagging and parsing are now often implicit inside a single model rather than separate steps you run.
- It made **IR** even more important. Even the best Large Language Models (LLMs) cannot hold every fact, so retrieval was reintroduced to supply them.
- It is entirely **ML**, at scale.

Most of what follows in this course is now a **component inside** a generative system rather than a pipeline stage beside it.



"Just pretend I'm not here..."

So Why Not Just Use an LLM?

Every task on the previous slides can be attempted, today, in one line:

- No training data. No feature engineering. No hassle.
- It works, in any language, on any domain, and it takes seconds to write.



is this review for "The Odyssey" positive or negative?

"It was good but the whole 21st century feel was offputting."

Mixed, but leaning positive. The reviewer liked it overall ("It was good") while criticizing the modern, 21st-century feel.



So why don't we study only Large Language Models?

How We Got Here

Three Eras

Rule-based

The earliest approach.

- Strict linguistic rules written by experts
- Large libraries of accurate rules
- Precise where it applies
- Fails on exceptions, and does not scale

Statistical

Learn from data instead.

- Models are fed examples
- Behavior is estimated from counts, not stipulated
- Needs annotated data
- Robust, but shallow

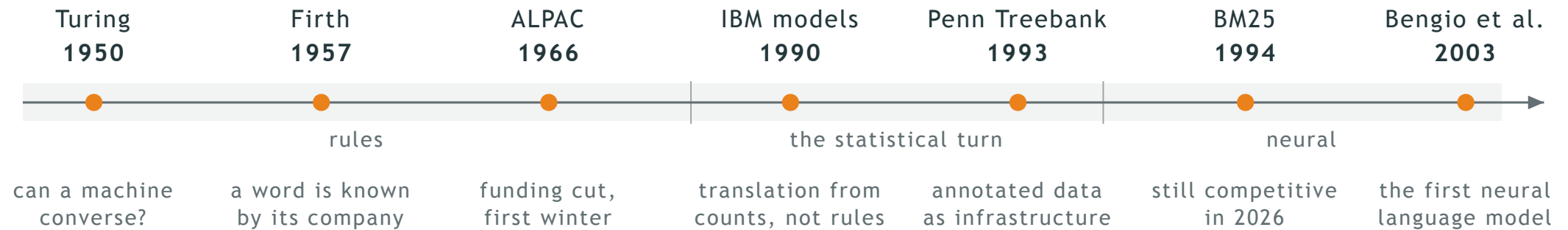
Neural

From roughly 2000.

- Representations are learned, not designed
- Enabled by data volume and parallel hardware (GPU, TPU)
- Very strong, very opaque

Historical Landmarks

The field is older than it looks.



What the Early Decades Established

1954, Georgetown-IBM. Sixty Russian sentences translated automatically, in public. The researchers predicted machine translation would be solved soon.

1957, Firth. *"You shall know a word by the company it keeps."* The **distributional hypothesis**: meaning can be recovered from context alone:

The bank was muddy today -- I had to wait one hour at the bank

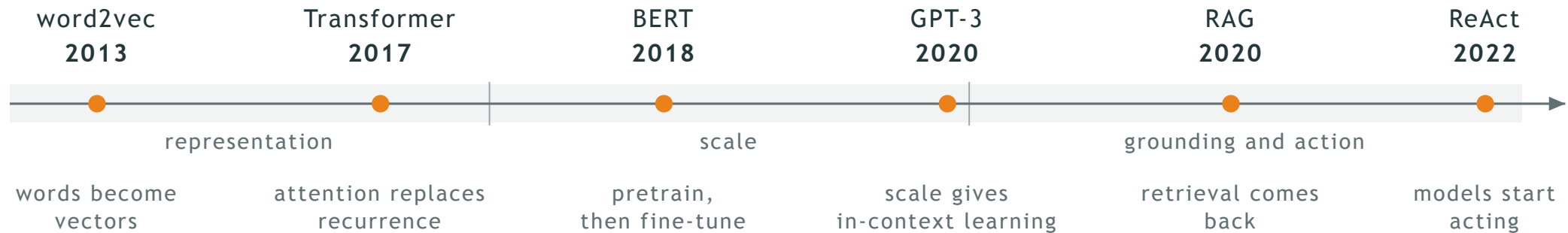
1966, ELIZA. A few pattern-matching rules produced conversations people found genuinely affecting. Weizenbaum was disturbed by how readily users attributed understanding to a program that had none.

1990 onward, the statistical turn. IBM's translation models learned from aligned text instead of grammars; the Penn Treebank made annotated data shared infrastructure; BM25 gave retrieval a weighting scheme.

Six Recent Landmarks

In nine years, the landscape moved:

- from representing words as vectors
- to building models that act.

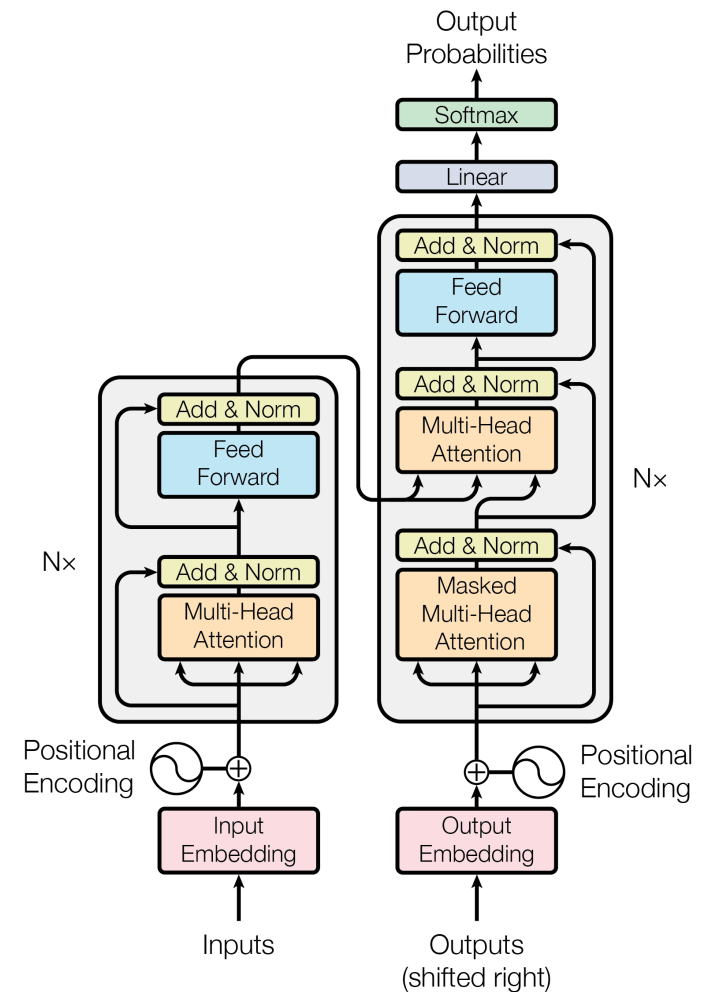


2013 to 2018: Representation

2013, word2vec (Mikolov et al.). Words become dense vectors learned from context, so that *similar* words land in *nearby* places. Similarity becomes geometry.

2017, Attention Is All You Need (Vaswani et al.). The Transformer drops recurrence entirely and processes a sequence in parallel, letting every position attend to every other. This is the architecture everything after it is built on.

2018, BERT (Devlin et al.). Pretrain a large model on unlabelled text, then fine-tune it on your small labeled task. The cost of a new application collapses, because you no longer start from nothing.



2020 to Today: Scale, Then Retrieval, Then Action



Gemini



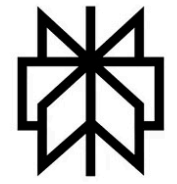
OpenAI



Meta



Grok



Perplexity



Apple
Intelligence



Claude

2020, GPT-3 (Brown et al., *Language Models are Few-Shot Learners*). At sufficient scale a model performs tasks it was never trained for, given only a description and a few examples. Fine-tuning stops being mandatory.

2020, RAG (Lewis et al.). Parameters are a poor place to keep facts: they go stale, they cannot be audited, and they cannot cite. So documents are **retrieved** at query time and given to the model.

2022, ReAct (Yao et al.) and the tool-use line. The model stops only answering and starts **acting**: calling tools, running searches, taking steps.

So Why Not Just Use an LLM?

1. LLMs have costs

The revolution is real: translation, summarization and question answering all changed. The limitations are equally real.

- **Data hungry.** Effective training needs very large datasets.
- **Opaque.** Decisions are hard to explain, which matters when a decision affects someone.
- **Expensive.** Time, memory and energy, at training and at inference.
- **Poor generalization** to rare or unseen concepts, which makes low-resource languages hard.
- **Bias.** Societal bias in the training data is reproduced and can be amplified.

None of these are solved.

LLMs are expensive

LLMs are expensive, but they are becoming less and less.

	cost per million tokens
November 2022, GPT-3.5 level	\$20.00
October 2024, same level	\$0.07

A 280-fold reduction in about 18 months. Hardware costs fall roughly 30% a year, energy efficiency improves roughly 40% a year, and inference prices for a given task have fallen between 9 and 900 times per year.

The AI Index Report attributes the cost collapse to one cause:

the fall in inference cost was driven by increasingly capable small models

The field did not get cheaper by making large models cheaper. It got cheaper by making **small, specialized** models good enough.

Specialization is not what the field is moving away from; it is what the field moved *toward* in order to become affordable.

Where the Field Is Going

A 2025 position paper from NVIDIA argues that small language models are *"sufficiently powerful, inherently more suitable, and necessarily more economical"* for agentic systems.

The reasoning is about what agents actually do:

- Real agents perform a **small number of specialized tasks, repetitively, with little variation**: intent classification, tool selection, structured output.
- On those tasks small models match or beat large ones, at **10 to 30 times lower inference cost**.
- The natural design is **heterogeneous**: a large model where open-ended conversation is genuinely needed, small specialized models everywhere else.

Choosing which is which is a judgment this course is meant to give you.

And Specialization Still Wins on Accuracy

Fine-tuned **small** models consistently outperform zero-shot prompting of much larger generative models on text classification: sentiment, stance, emotion.

- The comparison includes GPT-4 and Claude, prompted zero-shot.
- The small models need, on average, **10 to 1,000 labeled examples** to get there.
- They can be stored, trained and run **on a laptop**, which also settles the privacy question.

So the trade is not "cheap and worse" against "expensive and better". On a well-defined task with a little labeled data, the small specialized model is **cheaper and better**.

2. The Classics Are Inside the Modern Systems

It is worth knowing the classics also because they are used inside LLMs. Everything in the first half of this course reappears later as a component:

taught in	returns as
subword tokenization (BPE)	the tokenizer of every LLM
chain rule and n-grams	next-token prediction is $P(w \mid \text{context})$
perplexity	still the standard language-model metric
PMI	word2vec implicitly factorises a PPMI matrix
TF-IDF, cosine, the inverted index	retrieval used in RAGs
the vector space model	embedding space
precision and recall	how any of it is evaluated

3. Not everything is solved

In previous editions of this course, the introduction slides sorted tasks into *mature*, *progressing* and *not yet ready*:

	then	now
tagging, parsing, NER, search	mature	
translation, classification, clustering	progressing	
question answering, summarization, dialogue	not yet ready	
quantification, faithful attribution, calibrated uncertainty	not yet ready	

3. Not everything is solved

In previous editions of this course, the introduction slides sorted tasks into *mature*, *progressing* and *not yet ready*:

	then	now
tagging, parsing, NER, search	mature	mature , now often implicit inside a larger model
translation, classification, clustering	progressing	largely mature , specialized models still competitive
question answering, summarization, dialogue	not yet ready	usable , and the reason the public noticed the field
quantification, faithful attribution, calibrated uncertainty	not yet ready	still not ready

What Is Still Not Ready

The last row is the interesting one. **Fluency arrived long before reliability.** Three things a fluent model still does badly, and all three are measurement problems rather than writing problems:

- **Quantification.** *"What fraction of these reviews are negative?"* is not the same question as *"is this review negative?"*. A classifier that is 90% accurate can be badly wrong about the **proportion**, because its errors need not cancel.
- **Faithful attribution.** A system can cite a source, and the source can fail to support the claim. The citation looks right, reads right, and is not checked by anything in the model.
- **Calibrated uncertainty.** A confidence of 0.9 should be wrong one time in ten. Current models are confidently wrong, and the confidence they report is not a probability you can act on.

Fluency makes the failure no longer looks like a failure.

Text Is Rarely Alone

Human communication is **multimodal**, and text analytics increasingly borders on other data types:

- **Images and video.** Captioning describes visual content in words, which makes it searchable and accessible. Text about an image helps interpret the image, and the image helps interpret the text.
- **Tabular data.** Natural language questions over databases replace formal queries. In the other direction, extracting structure from historical text enables analysis of material never designed for it.
- **Graphs.** Social networks reveal structure and dynamics; adding the *content* of what is shared characterizes the network far better, and makes it possible to study manipulation and abnormal behavior.

What is the purpose of this course?

Suppose you accept that an LLM will do the work. You still have to:

- **Choose.** Large or small, prompted or fine-tuned, retrieval or parameters. Every one of these is a cost and quality trade-off you cannot evaluate without knowing what the alternatives do.
- **Evaluate.** A number without a metric is a claim without evidence.
- **Debug.** When retrieval returns nothing useful, the cause is usually vocabulary mismatch, not a prompting problem.
- **Explain.** To a colleague, a reviewer, or a regulator who is entitled to ask why a decision was made.

The aim of this course is not to prefer old methods to new ones or viceversa:

it is to make you the person in the room who can tell which one the problem needs.

The landmarks

- Mikolov et al., *Efficient Estimation of Word Representations in Vector Space*, 2013.
- Vaswani et al., *Attention Is All You Need*, 2017.
- Devlin et al., *BERT*, 2018.
- Brown et al., *Language Models are Few-Shot Learners*, 2020.
- Lewis et al., *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*, 2020.
- Yao et al., *ReAct: Synergizing Reasoning and Acting in Language Models*, 2022.

The numbers quoted in this lecture

- Stanford HAI, *AI Index Report 2025*, chapter 1, for inference cost.
- Belcak et al. (NVIDIA), *Small Language Models are the Future of Agentic AI*, 2025.
- Bucher & Martini, *Fine-Tuned 'Small' LLMs (Still) Significantly Outperform Zero-Shot Generative AI Models in Text Classification*, 2024.